

To calculate limits more explicitly, we utilize some Limit Laws to give us a guide on what we can and cannot do.

Limit Laws

Let c be a number and f, g functions s.t. $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ both exist.

$$1) \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$2) \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

$$3) \lim_{x \rightarrow a} [c f(x)] = c \lim_{x \rightarrow a} f(x)$$

$$4) \lim_{x \rightarrow a} [f(x) g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$5) \lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \text{if } \lim_{x \rightarrow a} g(x) \neq 0$$

$$6) \lim_{x \rightarrow a} [f(x)]^n = [\lim_{x \rightarrow a} f(x)]^n$$

$$7) \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}, \quad \text{if } n \text{ is even, } \lim_{x \rightarrow a} f(x) \geq 0$$

$$8) \lim_{x \rightarrow a} c = c$$

$$9) \lim_{x \rightarrow a} x = a$$

$$10) \lim_{x \rightarrow a} x^n = a^n$$

$$11) \lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}, \quad \text{if } n \text{ is even, need } a \geq 0$$

Some functions work very well with limits.

Direct Substitution Property

If f is a rational or polynomial function and a is in the domain of f , then

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Recall that polynomials have domain $(-\infty, \infty)$ and rationals have domain as all real numbers in which the denominator is non zero.

This helps for a lot of functions, but what about when a isn't in the domain. In this case there's another big result.

If $f(x) = g(x)$ when $x \neq a$, then $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$

provided the limits both exist.

$\boxed{=x}$

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x+1)(x-1)}{x-1} = \lim_{x \rightarrow 1} x+1 = 1+1 = \boxed{2}$$

let

$$g(x) = \begin{cases} x+1 & x \neq 1 \\ \pi & x = 1 \end{cases}$$

$$\lim_{x \rightarrow 1} g(x) = \lim_{x \rightarrow 1} x+1 = 1+1 = \boxed{2}$$

$$\lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h} = \lim_{h \rightarrow 0} \frac{\cancel{9} + 6h + h^2 - \cancel{9}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{6}h + h^{\cancel{2}}}{\cancel{h}}$$

$$= \lim_{h \rightarrow 0} 6+h = \boxed{6}$$

$$\lim_{t \rightarrow 0} \frac{\sqrt{t^2+9} - 3}{t^2} = \lim_{t \rightarrow 0} \frac{\sqrt{t^2+9} - 3}{t^2} \cdot \frac{\sqrt{t^2+9} + 3}{\sqrt{t^2+9} + 3}$$

$$= \lim_{t \rightarrow 0} \frac{t^{\cancel{2}} - \cancel{9}}{(t^{\cancel{2}})(\sqrt{t^2+9} + 3)}$$

$$= \lim_{t \rightarrow 0} \frac{\cancel{t}}{(\cancel{t})(\sqrt{t^2+9} + 3)} = \lim_{t \rightarrow 0} \frac{1}{\sqrt{t^2+9} + 3} = \boxed{\frac{1}{6}}$$

We can also use these limit laws when addressing sided limits. Specifically, recall that

$$\lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x) \quad \text{if and only if} \quad \lim_{x \rightarrow a} f(x) = L$$

We can then show

$$\lim_{x \rightarrow 0} \frac{|x|}{x} \text{ DNE, Note } |x| = \begin{cases} -x & x < 0 \\ x & x \geq 0 \end{cases}$$

So,

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = \lim_{x \rightarrow 0^-} -1 = -1$$

Since $x < 0$
as $x \rightarrow 0^-$

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = \lim_{x \rightarrow 0^+} 1 = 1$$

These are not equal, so $\lim_{x \rightarrow 0} \frac{|x|}{x} \text{ DNE}$

Ex:

$$f(x) = \begin{cases} \sqrt{x-4} & x > 4 \\ 8-2x & \underline{x \leq 4} \end{cases}$$


$$\lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^-} 8-2x = 0$$

$$\lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^+} \sqrt{x-4} = 0$$

$$\text{So, } \lim_{x \rightarrow 4} f(x) = 0$$